Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_no_replay.
- Best TSPLIB holdout gap: tsplib_no_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_134222_g
- started_at_local: 2026-05-13 13:42:22
- finished_at_local: 2026-05-13 13:51:55
- duration_hhmm: 00:10
- duration_seconds: 572.307
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.864896	0.56	-0.028795
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.872001	0.56	-0.028561
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.883365	0.56	-0.028193
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.864896, and final complexity 0.56.
- Adaptation efficiency -0.028795 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.872001, and final complexity 0.56.
- Adaptation efficiency -0.028561 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.883365, and final complexity 0.56.
- Adaptation efficiency -0.028193 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
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- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

Summary of Results for Replay-Aware TSP Benchmark Suite

Condition	Replay Mode	Adaptation Efficiency	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Compression Pressure	Comments
tsplib_no_replay	none	-0.028795	0.213387	0.064916	0.159398	0.864896	no	Best transfer and gaps. Highest code novelty among replay conditions.
tsplib_random_replay	random	-0.028561	0.213387	0.064916	0.159398	0.872001	no	Identical performance to no replay despite slightly lower novelty.
tsplib_failure_replay	failure	-0.028193	0.213387	0.064916	0.159398	0.883365	no	Same transfer and optimality gaps as no replay, with marginally higher code novelty than no replay condition.
tsplib_failure_replay_compression	failure + compression	0.0	0.213387	0.064916	0.159398	0.0	yes	Identical transfer and gaps but zero code novelty due to compression-aware replay and novelty gating which limited changes.

Conservative Interpretation:

- **Optimality Gaps & Transfer Evidence:**

Across all four conditions, the **mean TSPLIB gap (~0.2134)**, **synthetic gap (~0.0649)**, and **transfer gap (~0.1594)** are exactly the same, indicating no measurable improvement or degradation under any replay regime in this benchmark suite.

- **Replay Modes:**
- *No replay (tsplib_no_replay)* performed equally well and is identified as the best condition overall.
- *Random replay* and *failure replay* did not improve transfer or gaps relative to no replay.
- *Compression-aware failure replay* with novelty gating showed a collapse in code novelty (mean = 0), but did not improve transfer outcomes. This suggests that increased compression pressure and novelty gating limited code changes without positively impacting transfer or optimality gaps.

- **Code Novelty vs Transfer:**

Although the *tsplib_failure_replay* condition had slightly higher mean code novelty (0.883) than *no replay* (0.865), the transfer and gap results were identical. Hence, increased code novelty does **not** translate here into better algorithmic generalization or transfer performance.

Conversely, the compression-aware replay condition drastically reduces code novelty to zero but does not outperform other conditions; transfer metrics remain matched.

- **Algorithmic Novelty:**

Given that all optimality gaps and transfer metrics are identical across conditions, **none of the replay modes provided algorithmic improvements**. The lexical/code novelty metric variation alone is insufficient to claim algorithmic invention.

- **Archived Archives and Behavior:**

All conditions maintain balanced final behavior profiles and similar elite/total archive sizes, with no compression pressure except in the compression-aware replay.

Conclusion:

- The **no replay condition** currently provides the best empirical evidence for transfer effectiveness, showing minimal gaps and good transfer without replay complexity.
- Replay modes (random, failure) do not improve transfer or final optimality gaps compared to no replay, despite minor differences in code novelty.
- Compression-aware replay with novelty gating suppresses code novelty but does not improve transfer.
- Optimality gap and transfer metrics provide no support for claims that replay mechanism variations yield better performance in this benchmark.
- Code novelty differences among conditions do not correlate with improved transfer results and should not be interpreted as algorithmic advancement.

Recommendations:

Prioritize the no replay baseline in further development until replay modes demonstrate improved transfer or reduced optimality gaps. Code novelty alone is insufficient evidence of algorithmic invention without corresponding improvements in deterministic metrics.